

Problem 1: Put 6 counters in piles of 4, 1 and 1, like in the launch. A move goes like this: take one counter from every pile, and put those counters together as one new pile. Make one move. In row 1 of the table, write the sizes of your piles, biggest first, like this: 3, 3. Keep making moves and writing a row for each one, until a row is exactly the same as the row above it. When that happens, the game has stopped. How many moves did it take? If you run out of rows in any table, continue on a blank sheet.

Move	Piles, biggest first
Start	4, 1, 1
1	
2	
3	
4	
5	
6	
7	
8	

Moves until it stopped: _____

Problem 2: Do the same from two more starts: all 6 counters in one pile, and six piles of 1. Use one table for each start. Keep making moves until a row is the same as an earlier row. If it is the same as the row just above it, the game has stopped.

Move	Piles, biggest first
Start	6
1	
2	
3	
4	
5	
6	
7	
8	

Move	Piles, biggest first
Start	1, 1, 1, 1, 1, 1
1	
2	
3	
4	
5	
6	
7	
8	

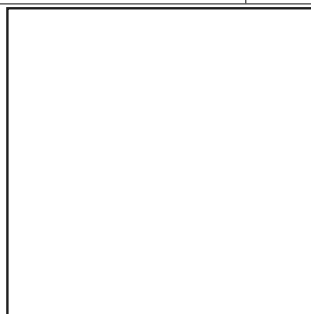
Problem 3: Do the same from two more starts: two piles of 3, and piles of 2, 2, 1 and 1. Keep making moves until a row is the same as an earlier row.

Move	Piles, biggest first
Start	3, 3
1	
2	
3	
4	
5	
6	
7	
8	

Move	Piles, biggest first
Start	2, 2, 1, 1
1	
2	
3	
4	
5	
6	
7	
8	

Problem 4: Fill in the table for your five starts: did the game stop or go round, after how many moves, and what were the piles at the end? Then build the end piles as towers standing side by side, biggest on the left, and draw the towers in the box.

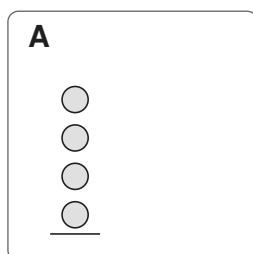
Start	Stopped, or went round?	Moves	Piles at the end
4, 1, 1			
6			
1, 1, 1, 1, 1, 1			
3, 3			
2, 2, 1, 1			



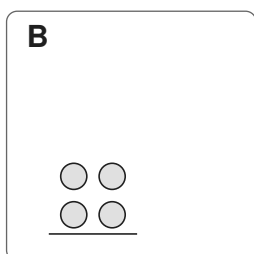
Problem 5: Take 4 counters in one pile. Make 6 moves and write a row for each move. Circle every row that is the same as row 1. Does this game ever stop changing?

Move	Piles, biggest first
Start	4
1	
2	
3	
4	
5	
6	
7	
8	

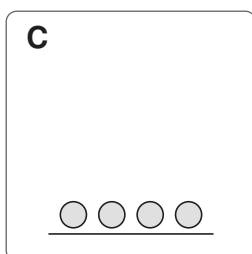
Problem 6: Here are all five ways to put 4 counters in piles, as pictures with letters. For each picture: make it with your counters, make one move, and write the letter of the picture you get in the blank. Then answer: which letters does the game go round and round on? Which picture can you never get by a move from another one?



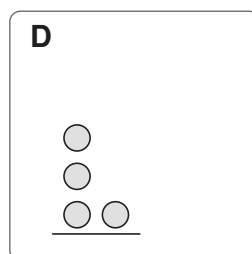
→ _____



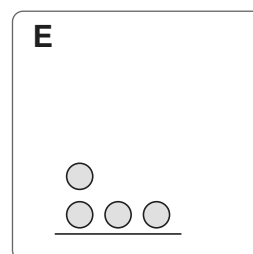
→ _____



→ _____



→ _____



→ _____

Goes round on: _____

Never reached by a move: _____

Problem 7: Take 7 counters in one pile. Make moves and write each row, until a row is the same as an earlier row. Circle the two rows that are the same. Does this game stop, or go round? If it goes round, how many different arrangements does it go round on?

Move	Piles, biggest first
Start	7
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

Stops, or goes round: _____

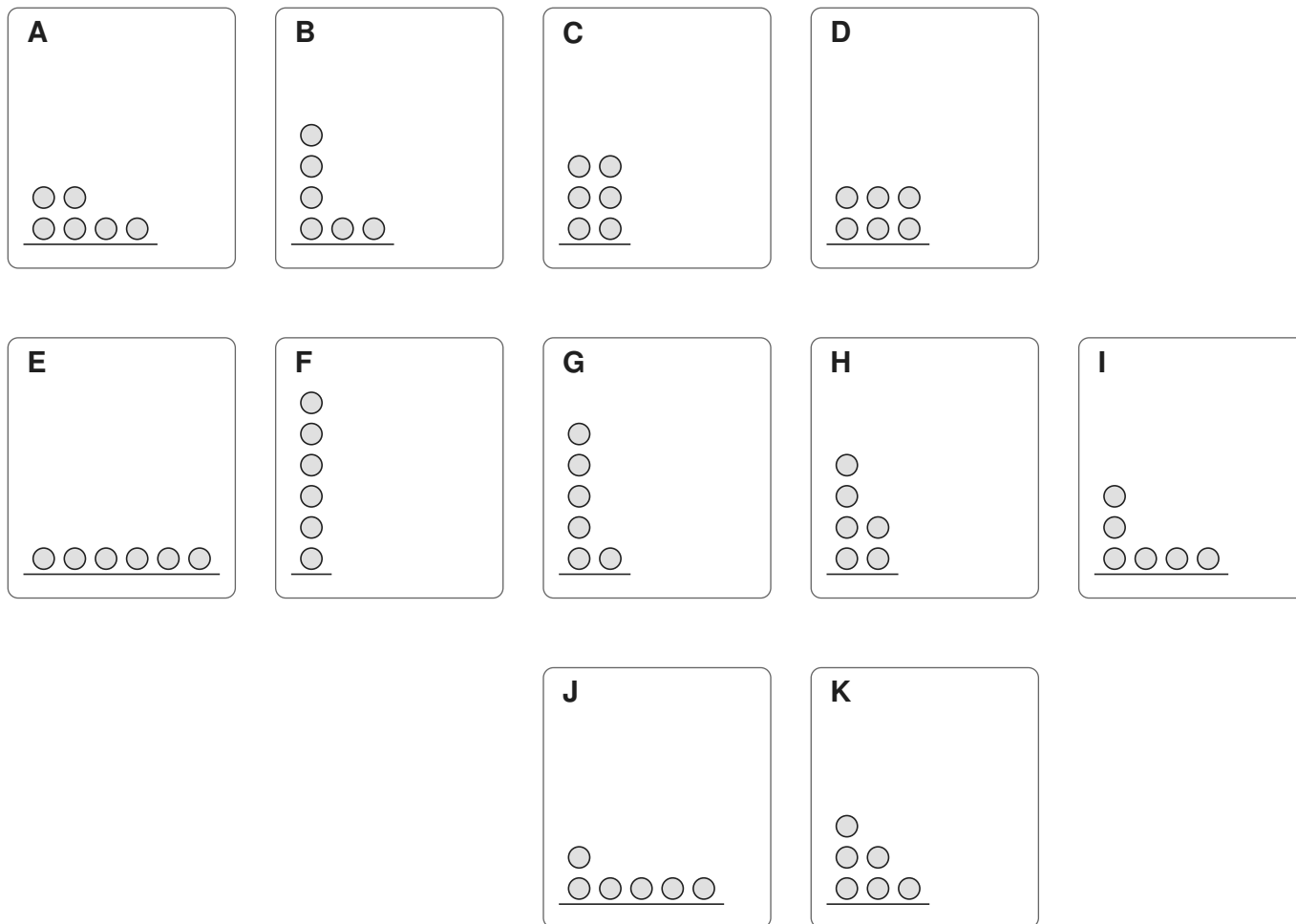
Arrangements in the loop: _____

Problem 8: Run 3 counters: start with three piles of 1 and make moves until a row is the same as an earlier row. Then fill in the table for 3, 4, 6 and 7 counters: does the game stop, or go round and round? If it goes round, how many different arrangements are in the loop? The last row is for another number you try.

Move	Piles
Start	1, 1, 1
1	
2	
3	
4	
5	

Counters	Stops, or goes round?	Arrangements in the loop
3		
4		
6		
7		

Problem 9: Here are all 11 ways to put 6 counters in piles, drawn as towers, with letters A to K. For each picture: make it with your counters, make one move, find the picture of the result, and draw an arrow from the first picture to the second. If the result is the same picture, draw a little arrow that loops back to it. Then answer the three questions under the map.



Which pictures have no arrow pointing at them? _____

Which picture do all the arrows lead to in the end? _____ Starting at A, how many arrows do you follow to get there? _____

Problem 10: The game with 3 counters stopped at towers of 2, 1. With 6 counters it stopped at towers of 3, 2, 1. Build both shapes. Then build the next stairs, one step taller, and count the counters. Write the number in the first blank. Write your prediction in the second blank: will the game with that many counters stop, or go round? Then test it: put that many counters in one pile, write the start in the table, and keep making moves until a row is the same as an earlier row.

Counters in the next stairs: _____ Prediction: _____

Move	Piles, biggest first
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	

What happened: _____

After how many moves: _____

Problem 11: Now 8 counters. Put all 8 in one pile and make moves, writing each row, until a row is the same as an earlier row. Then put 8 counters in four piles of 2 and do the same. In each table, circle the rows that are the same. What happens in each game: does it stop, or go round? If both go round, do they go round on the same arrangements?

Move	Piles, biggest first
Start	8
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

Move	Piles, biggest first
Start	2, 2, 2, 2
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

From 8: _____ From 2, 2, 2, 2: _____

Same arrangements in both loops? _____

Problem 12: Take 5 counters in one pile and make moves, writing each row, until a row is the same as an earlier row. Then start with five piles of 1 and do the same. In each table, circle the rows that are the same. Do both games go round on the same arrangements? Fill in the last row of the table in Problem 8 for 5 counters.

Move	Piles, biggest first
Start	5
1	
2	
3	
4	
5	
6	
7	
8	

Move	Piles, biggest first
Start	1, 1, 1, 1, 1
1	
2	
3	
4	
5	
6	
7	
8	

Same arrangements in both loops? _____

Problem 13: Run 10 counters from two more starts: five piles of 2, and two piles of 5. Keep making moves until a row is the same as an earlier row. For each start, write what happened and after how many moves.

Move	Piles, biggest first
Start	2, 2, 2, 2, 2
1	
2	
3	
4	
5	
6	
7	
8	
9	

Move	Piles, biggest first
Start	5, 5
1	
2	
3	
4	
5	
6	
7	
8	
9	

From 2, 2, 2, 2, 2: _____

From 5, 5: _____

Problem 14: Here are all 22 ways to put 8 counters in piles. Share them out at your table. For each one, run the game until a row is the same as an earlier row, and write 4 or 2 in the blank: 4 if it goes round on four arrangements, 2 if it goes round on two. If your run reaches an arrangement someone has already run, stop: you know where it ends. How many of the 22 go round on two?

8	_____	3, 3, 2	_____
7, 1	_____	3, 3, 1, 1	_____
6, 2	_____	3, 2, 2, 1	_____
6, 1, 1	_____	3, 2, 1, 1, 1	_____
5, 3	_____	3, 1, 1, 1, 1, 1	_____
5, 2, 1	_____	2, 2, 2, 2	_____
5, 1, 1, 1	_____	2, 2, 2, 1, 1	_____
4, 4	_____	2, 2, 1, 1, 1, 1	_____
4, 3, 1	_____	2, 1, 1, 1, 1, 1, 1	_____
4, 2, 2	_____	1, 1, 1, 1, 1, 1, 1, 1	_____
4, 2, 1, 1	_____		
4, 1, 1, 1, 1	_____		

Number that go round on two arrangements: _____

Problem 15: Build the stairs with 5 steps. Count the counters and write the number in the blank. Then put that many counters in five piles of 3. Make moves and write each row until a row is the same as an earlier row. What happened, and after how many moves?

Counters in the stairs with 5 steps: _____

Move	Piles, biggest first
Start	3, 3, 3, 3, 3
1	
2	
3	
4	
5	
6	
7	
8	

What happened: _____

After how many moves: _____